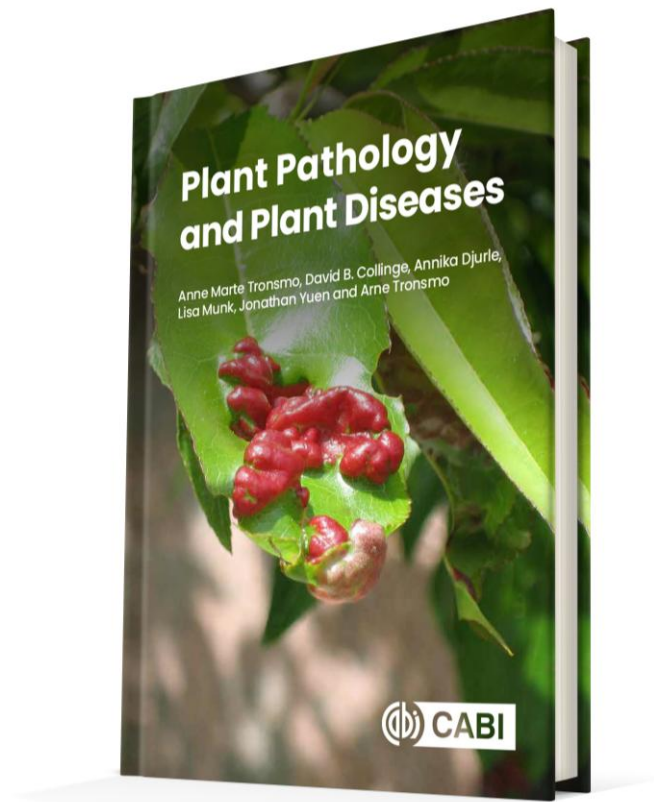




Plant Pathology and Plant Diseases

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Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

Fig. 5.1 Characteristic structures for oomycetes. (A) The hyphae lack transverse walls. (B) Zoosporangium. (C) Biflagellate zoospores. (D) Fertilization of an oogonium by antheridia; and oospore. (© H. Karlsen.)

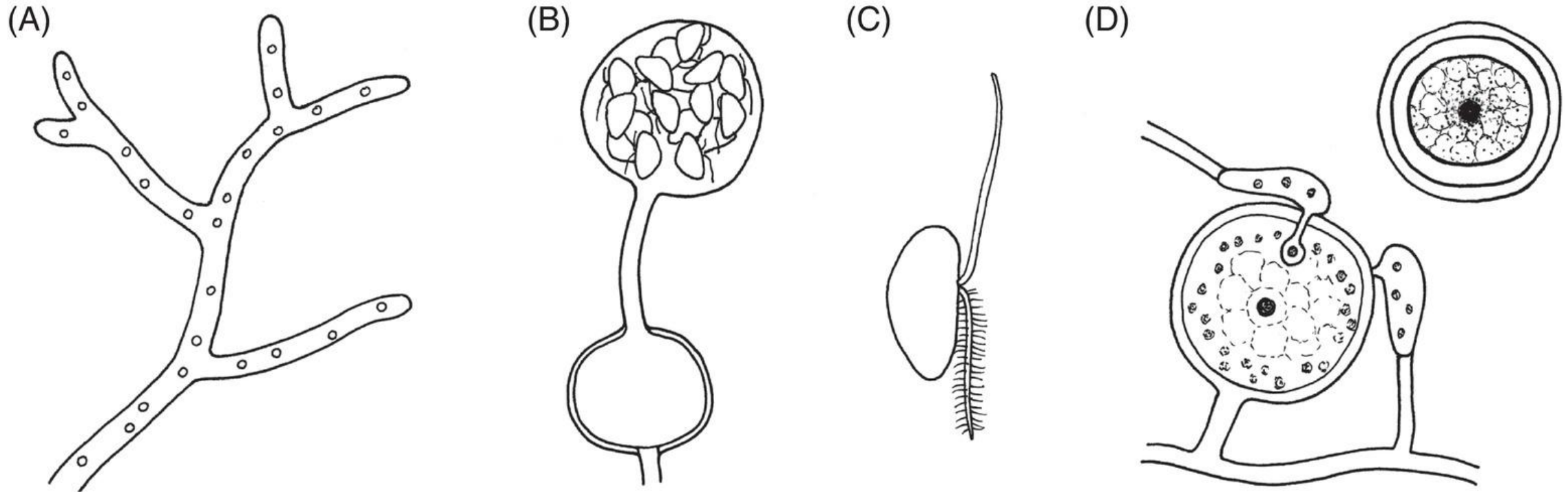


Fig. 5.2 Life cycle of an oomycete (*Pythium* sp.). (© H. Karlsen.)

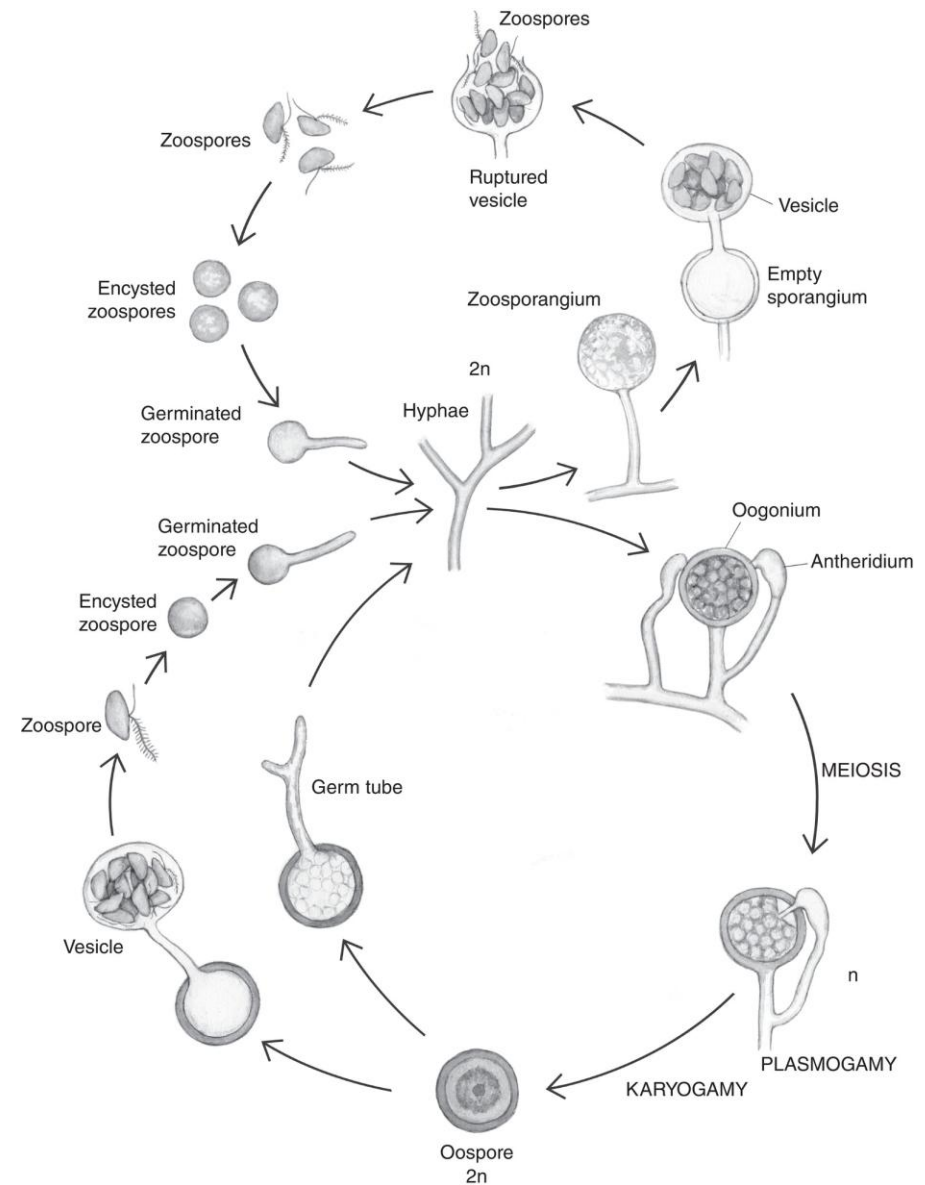


Fig. 5.3 Sexual and asexual structures of genera in Peronosporae. (A) *Albugo candida*. (B) *Phytophthora fragariae* var. *fragariae*. (C) *Bremia lactucae*. (D) *Plasmopara viticola*. (E) *Pythium* (left: *P. ultimum*; right: *P. oligandrum*). (F) *Aphanomyces euteiches*. (N. Leroul, courtesy of J. Hockenhull et al.)

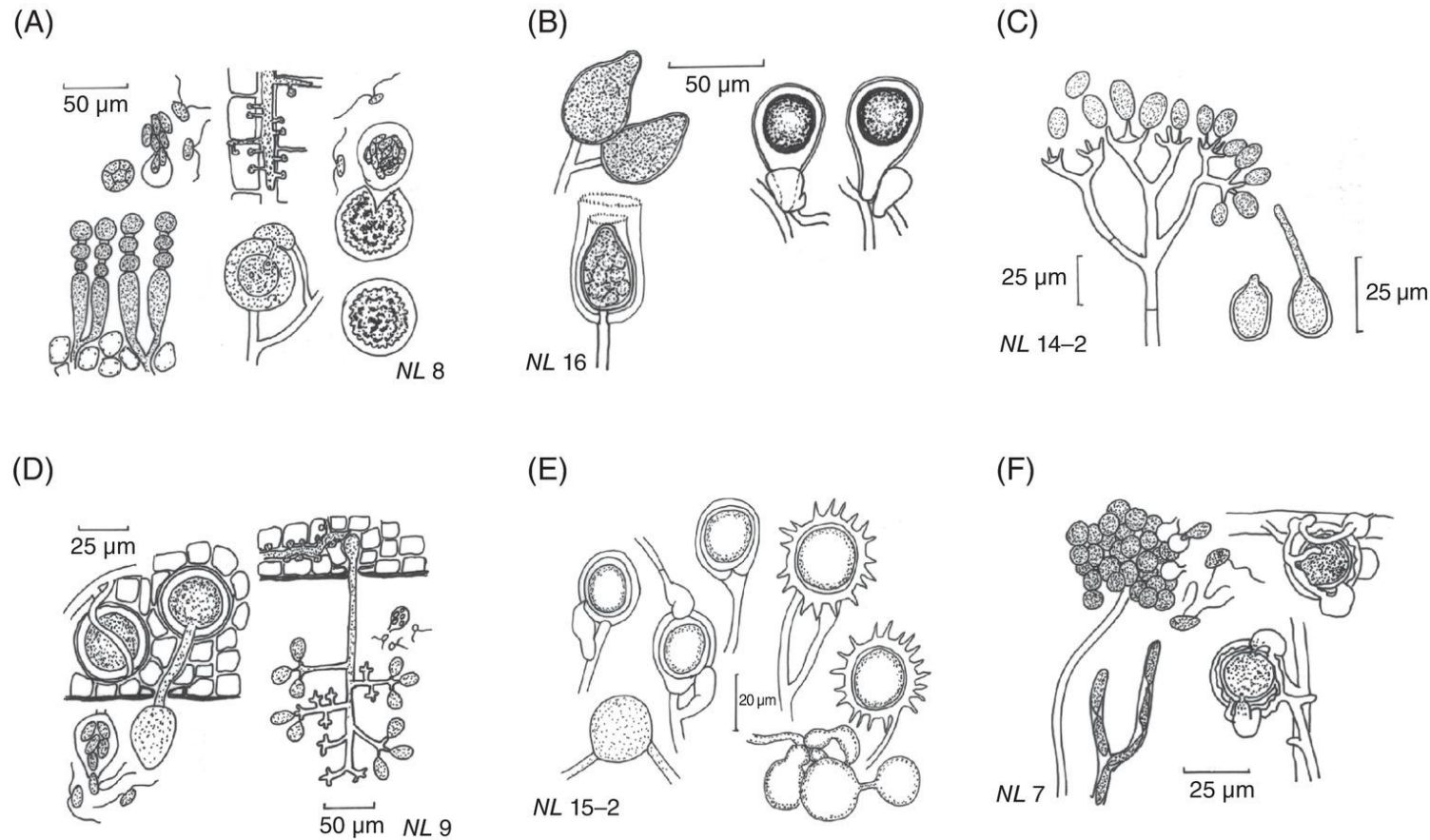


Fig. 5.4 A-B *Phytophthora infestans* symptoms and signs on potato. (A) Necrotic tissue on potato stalk. (B) Potato leaf with sporulating *P. infestans*. (© B. Anderson.)



Fig. 5.4 C-D *Phytophthora infestans* symptoms and signs on potato. (C) Sporangia of *P. infestans*. (D) An oospore of *P. infestans*. (© B. Anderson.)



Fig. 5.4 E-F *Phytophthora infestans* symptoms and signs on potato. (E) Infected potato tuber. (F) Transverse section of an infected potato. (E-F: E. Fløistad © NIBIO.)



TEACHING MATERIALS



Fig. 5.6 Ramorum dieback caused by *Phytophthora ramorum* in rhododendron. (A) Bud and (B) stems and leaves.
(A: © B. Toppe; B: E. Fløistad © NIBIO.)



Fig. 5.7 Symptoms of red core in strawberry caused by *Phytophthora fragariae*. (H.M. Singh © NIBIO.)



Fig. 5.8 Downy mildews. (A) *Bremia lactucae* on lettuce. (B) *Hyaloperonospora parasitica* on *Brassica oleracea* seedlings. (C) *Peronospora destructor* on onions. (D) *Plasmopara viticola* on vine. (A: E. Fløistad © NIBIO; B: B. Dahl Jensen © NIBIO; C: A. Hermansen © NIBIO; D: © L. Munk.)



Fig. 5.9 Disease cycle of downy mildew (*Peronospora destructor*) on onion.
(H. Karlsen © NIBIO.)

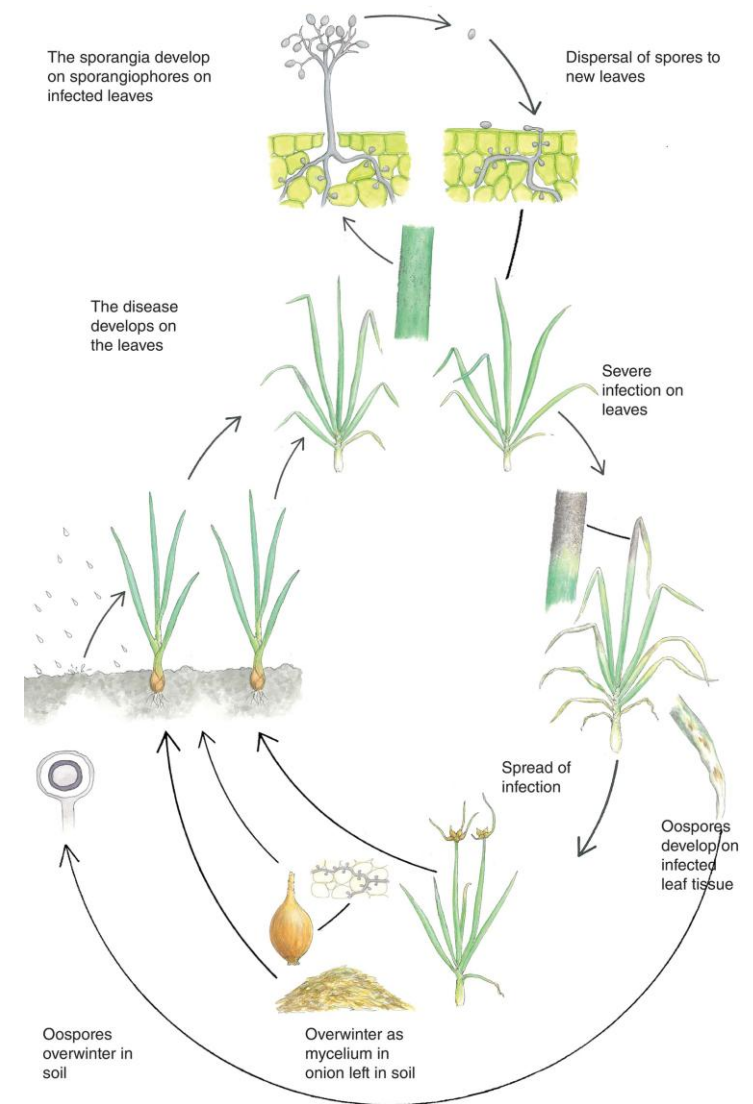


Fig. 5.10 Diseases caused by *Pythium*. (A) Pythium root rot, caused by *P. aphanidermatum*, in cucumber. (B) Shoulder rot, caused by *P. tracheiphilum* in Chinese cabbage. (A: E. Fløistad © NIBIO; B: A. Hermansen © NIBIO.)



Fig. 5.11 (A) Pea plants infected with *Aphanomyces euteiches* (left) and healthy pea plants right). (B) Pea plants showing varying degree of root injury after infection with *Aphanomyces euteiches*. (A-B: L. Sundheim © NIBIO.)



Fig. 5.12 A-B Clubroot symptoms in: (A) Swede and (B) Chinese cabbage. (A-B: E. Fløistad © NIBIO).



Fig. 5.12 C-D Clubroot symptoms (C) *Plasmodiophora brassicae* structures: clockwise from top left: plasmodium in root cells, zoospore, sporangium with zoospores, resting spores being released from host cell, resting spores, and a single zoospore (primary zoospore) developed from a resting spore. (D) Electron micrograph of resting spores of *Plasmodiophora brassicae* in host cells. (C: N. Leroul, courtesy of J. Hockenhull et al.; D: K. Brismar and S. Marttila, © SLU.)

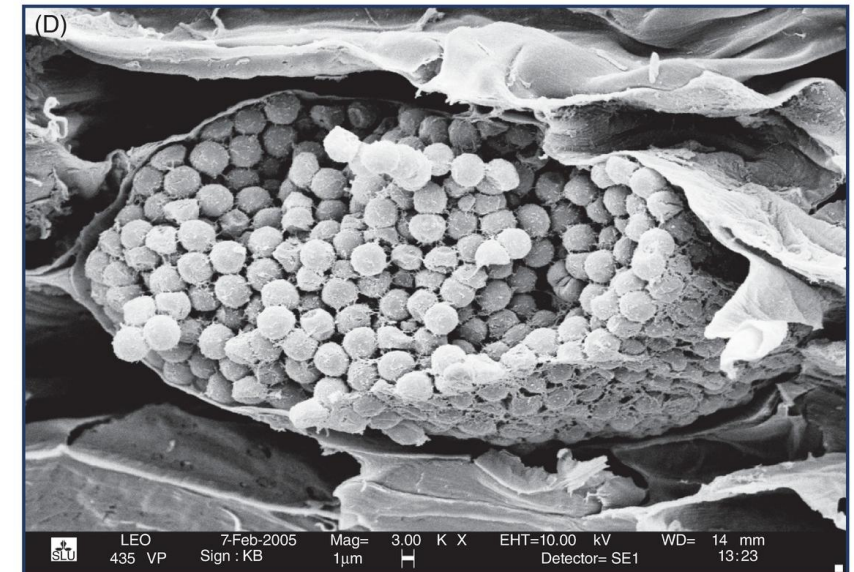
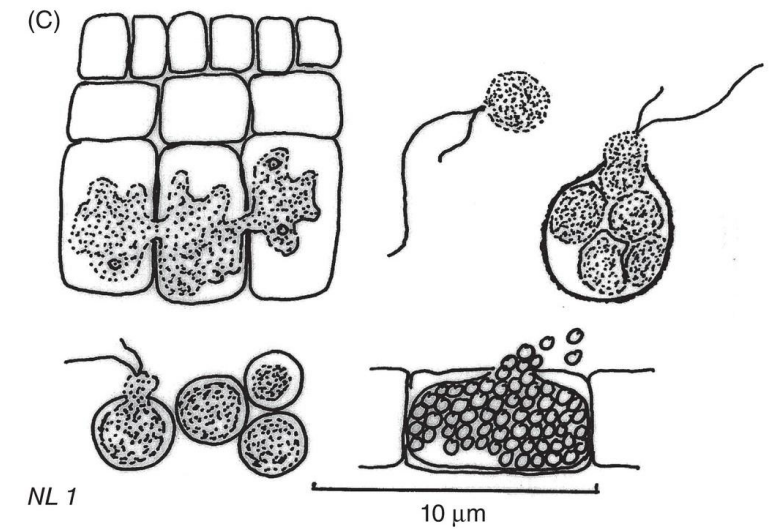


Fig. 5.13 *Spongospora subterranea*. (A) Symptoms and signs ('powder' of spores) on a potato tuber. (B) Resting spores in an almost globose mass and zoospores. (A: E. Fløistad © NIBIO; B: N. Leroul, courtesy of J. Hockenhull et al.)

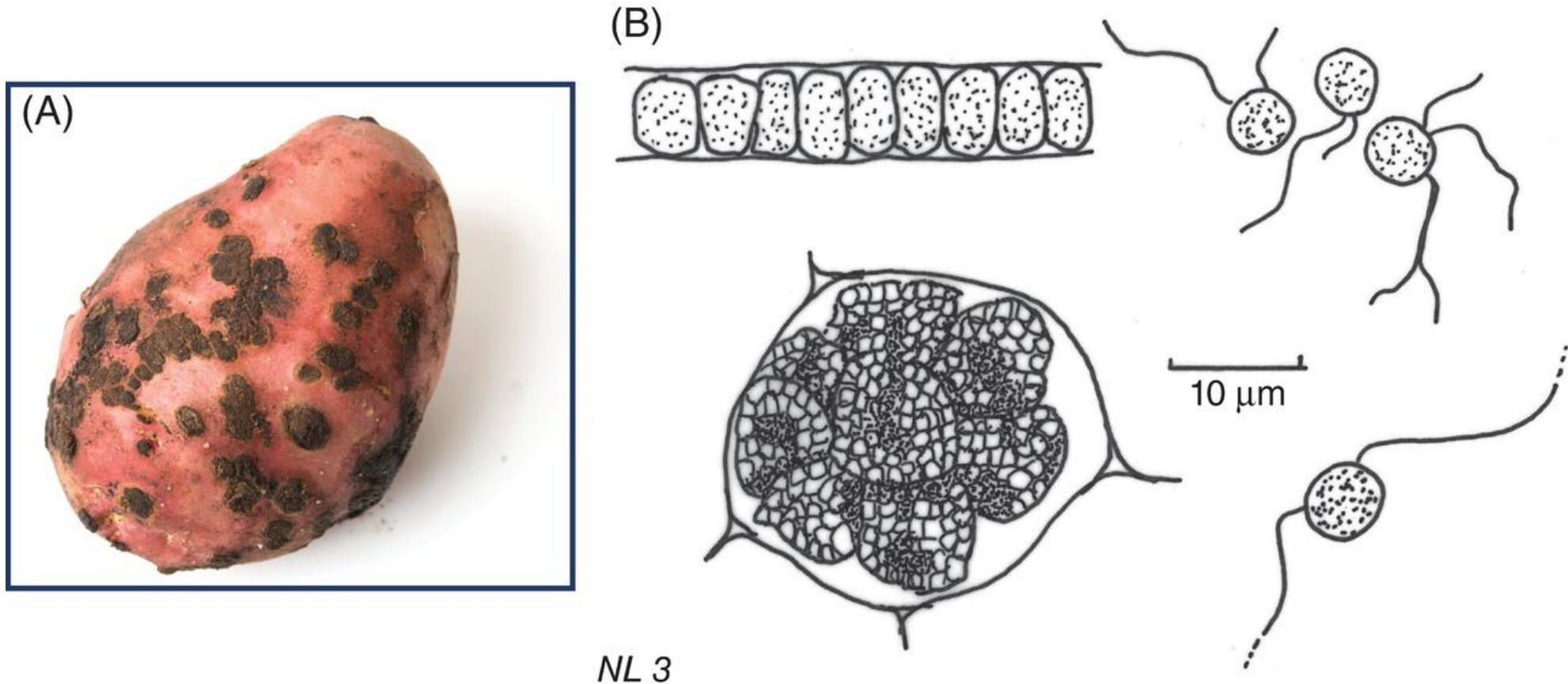


Fig. 5.14 Disease cycle of *Plasmodiophora brassicae*. (H. Karlsen © NIBIO.)

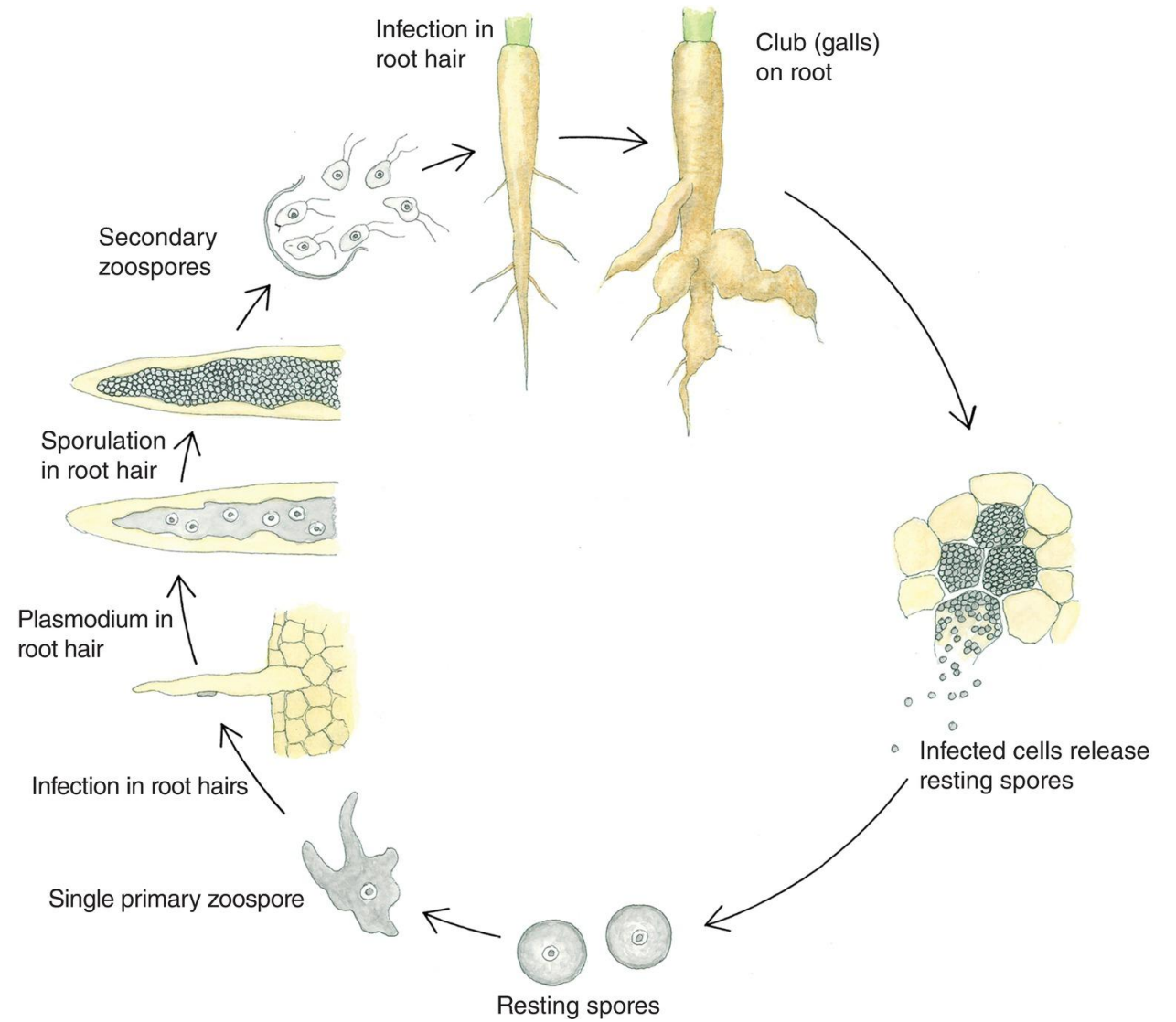


Fig. 5.15 Slime mould on turf grasses. (R. Lagnes © NIBIO.)

